

File Formats & Public Databases in Bioinformatics

[!IMPORTANT]Read the theory sections below for further practice and reference, during class we will not have time to read the entire document.

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Public repositories:

There are three main sites that have been responsible for storing **nucleotide sequence** data from 1982 to the present and are all coordinated by the International Nucleotide Sequence Database Collaboration (INSDC):

1. [GenBank at the National Center for Biotechnology Information \(NCBI\) of the National Institutes of Health \(NIH\)](#)
2. [European Molecular Biology Laboratory \(EMBL\)-Bank Nucleotide Sequence Database \(EMBL-Bank\)](#)
3. [DNA data Bank from Japan DDBJ](#)

	NCBI	EMBL	DDBJ
Primary sequence database	GenBank	ENA (European Nucleotide Archive)	DDBJ (same name as the organization)
Founded	1982 (GenBank)	1980 (EMBL Data Library); ENA name since 2008	1986
Metadata File Format*	GenBank format (.gb)	EMBL format (.embl)	similar to GenBank
Specialized Genome Browser**	Genome Data Viewer	Ensembl	DDBJ Genome Browser
Protein database	NCBI protein database	Uniprot	DDJB protein database
Curation	reference sequences RefSeq: NCBI-specific	NA	NA

***Metadata file format:**

This an example for the human major histocompatibility complex gene metadata. Notice how the exact same biological biological details (the exons, the protein sequence, and the organism) are structuralized using two completely different naming formats.

GeneBank format

```

LOCUS   NM_002116       3515 bp  mRNA  linear  PRI 09-AUG-2026
DEFINITION Homo sapiens major histocompatibility complex, class I, A (HLA-A),
            transcript variant 1, mRNA.
ACCESSION NM_002116
VERSION   NM_002116.8
KEYWORDS  RefSeq.
SOURCE    Homo sapiens (human)
            ORGANISM Homo sapiens
                Eukaryota; Metazoa; Chordata; Mammalia; Primates; Hominidae; Homo.
FEATURES             Location/Qualifiers
     source             1..3515
                        /organism="Homo sapiens"
                        /mol_type="mRNA"
                        /db_xref="taxon:9606"
     gene               1..3515
                        /gene="HLA-A"
                        /gene_synonym="HLAA"
                        /db_xref="GeneID:3105"
     CDS                73..1170
                        /gene="HLA-A"
                        /codon_start=1
                        /product="HLA class I histocompatibility antigen, A-3 alpha chain"
                        /protein_id="NP_002107.4"
                        /translation="MAVMAPRTL L L L L LSGALALTQTWAGSHSMRYFFTSVSRPGRGEPR
FIAVGYVDDTQFVRFDSDAASQRMEPRAPWIEQEGPEYWDQETRNVKAQSQTDRVDLGT
LRGYNQSEAGSHTIQIMYGCDVGS DGRFLRGYRQDAYDGKDYIALNEDLRSWTAADMA
AQITKRKWEAVHAAEQRRVYLEGRCDGLRRYLENGKETLQRTDPPKTHMTHHPISDHE
ATLRCWALGFYPAEITLTWQRDGEDQTQDTELVETRPAGDGTGFKWAAVVVPSGEEQRY
TCHVQHEGLPKPLTLRWELSSQPTIPVGIAGLVLLGAVITGAVVAAMWRRKSSDRK
GGSYSQAASSDSAQGS DVS LTACKV"
ORIGIN
1 gaggccgcgc cgacccgcgc gggccgcggg gggccggggc gaccgccgcg ggtactgagg
61 cccccgcgat ctgtgatggc cgtcatggcg ccccgaaacc tcctcctgct actctcgggg
121 gccctggccc tgaccagac ctgggcgggg tctactcca tgaagtatta ttcacctcc
//

```

EMBL format

```

ID  NM_002116; SV 8; linear; mRNA; STD; PRI; 3515 BP.
XX
DE  Homo sapiens major histocompatibility complex, class I, A (HLA-A),
DE  transcript variant 1, mRNA.
XX
AC  NM_002116;
XX
KW  RefSeq.
XX
OS  Homo sapiens (human)
OC  Eukaryota; Metazoa; Chordata; Mammalia; Primates; Hominidae; Homo.
XX
FH  Key      Location/Qualifiers
FH

```

```

FT source      1..3515
FT             /organism="Homo sapiens"
FT             /mol_type="mRNA"
FT             /db_xref="taxon:9606"
FT gene        1..3515
FT             /gene="HLA-A"
FT             /gene_synonym="HLAA"
FT             /db_xref="GeneID:3105"
FT CDS         73..1170
FT             /gene="HLA-A"
FT             /codon_start=1
FT             /product="HLA class I histocompatibility antigen, A-3 alpha chain"
FT             /protein_id="NP_002107.4"
FT             /translation="MAVMAPRTL L L L L L S G A L A L T Q T W A G S H S M R Y F F T S V S R P G R G E P R
FT             F I A V G Y V D D T Q F V R F D S D A A S Q R M E P R A P W I E Q E G P E Y W D Q E T R N V K A Q S Q T D R V D L G T
FT             L R G Y Y N Q S E A G S H T I Q I M Y G C D V G S D G R F L R G Y R Q D A Y D G K D Y I A L N E D L R S W T A A D M A
FT             A Q I T K R K W E A V H A A E Q R R V Y L E G R C V D G L R R Y L E N G K E T L Q R T D P P K T H M T H H P I S D H E
FT             A T L R C W A L G F Y P A E I L T W Q R D G E D Q T Q D T E L V E T R P A G D G T F Q K W A A V V V P S G E E Q R Y
FT             T C H V Q H E G L P K P L T L R W E L S S Q P T I P V G I I A G L V L L G A V I T G A V V A A V M W R R K S S D R K
FT             G G S Y S Q A A S S D S A Q G S D V S L T A C K V"
XX
SQ Sequence 3515 BP; 712 A; 1045 C; 1083 G; 675 T; 0 Other;
   gaggcgcgcg cgaccgcgcg gggccgcggg gggccggggc gaccgccgcg ggtactgagg   60
   ccccgcatg ctgtgatggc cgctatggcg ccccgaaacc tcctctgct actctcgggg   120
   gccctggccc tgaccagac ctgggcgggg tctcactcca tgaagtatta ttcacctcc   180
//

```

[!NOTE] :question: When writing you do you think it is important to know the differences between the two file formats

****Genome browser:** These are interactive, graphical user interfaces that allows us to explore the blueprint of an organism's DNA. The most popular genome browsers are NCBI, Ensembl and UCSC, the main differences between them are:

	NCBI	Ensembl	UCSC
Run by	US government (NIH/NLM)	EMBL-EBI + Wellcome Sanger (UK/EU)	UC Santa Cruz
Core identity	The main general purpose archive: sequences,	Genome annotation + comparative genomics across	Genome browser + annotation, strong human/model-

	literature, bioprojects	species	organism focus
Accession labels	NM_, NR_, XM_, XR_ (RefSeq)	Stable IDs: ENSG (gene), ENST (transcript), ENSP (protein)	Often reuses RefSeq/GenBank IDs

Genome browsers display many categories of information about chromosomal features, including genes, regulatory regions, variation, introns, exons, mRNA, non-coding DNA, isoforms, etc.

The basic formats for high-throughput biological data

There are multiple formats to storage, represent and analyze biological data. As you start creating your own scripts, workflows or if you ended up developing your own bioinformatic tools, **understanding the exact structure and specific information contained within each format is crucial**.

These are the most common formats that you will work with.

Format	Stores	Human-readable?	Typical use	Example tool/database
GenBank flat file (.gb, .gbk)	Sequence + rich embedded annotation in one file	Yes (plain text)	Single-record view combining sequence and feature annotation	NCBI
FASTA (.fa, .fasta)	Raw sequence only (DNA, RNA, or protein), no quality, no annotation	Yes (plain text)	Reference genomes, gene/protein sequences	NCBI, Ensembl, UCSC

<u>FASTQ</u> (.fastq, .fq)	Raw sequencing reads + per-base quality scores	Yes (plain text)	Direct output of a sequencing machine, pre-analysis	SRA, ENA/ DDBJ (DRA)
<u>GTF</u> (.gtf)	Gene annotation, it contains coordinates, feature type, strand, key-value attributes	Yes (plain text)	Annotate and locate genes in a genome/ transcriptome	NCBI RefSeq, Ensembl
<u>GFF3</u> (.gff, .gff3)	Gene annotation, similar to GTF, hierarchical structure	Yes (plain text)	Same purpose as GTF, different syntax/ structure	NCBI, Ensembl
<u>BED</u> (.bed)	Simple genomic intervals: chromosome, start, end (+ optional name/ score/strand)	Yes (plain text)	coordinates, regions of interest, custom feature sets	Created by UCSC
<u>SAM</u> (.sam)	Sequence Alignment Map, reads aligned to a reference genome, with position + alignment quality	Yes (plain text)	Human-readable version of alignment data	Output of aligners (BWA, Bowtie, STAR)
<u>BAM</u> (.bam)	Same data as SAM but binary compressed	No (binary)	Efficient storage/ processing of large alignment datasets	Same as SAM

Practice:

Today's organism: *Escherichia coli* str. K-12 substr. MG1655 this is the classic reference bacterium in molecular biology because it has a small genome (~4.6 Mb), is fully annotated, and available from both NCBI and Ensembl.

Step 1 — Set up your working directory for today

[!CAUTION]MAC USERS, MAKE YOUR TERMINAL BASH

You already have your btec_640 directory in your own computer.

1. Create today's **working directory**, inside your `/btec_640/class_exercices/` directory. Remember that inside your **working directory** you need to create `input_data` and `analysis` directories. Do it using a single command line.

🔗 [Click here for a hint](#) `mkdir -p workdir/input_data workdir/analysis`

#Paste your command here:

```
Quant@WIN-KPSE31J44DI MINGW64 ~/Documents/GitHub/BTEC_640/exercises/09_21_26 (main)
$ mkdir input_data
```

```
Quant@WIN-KPSE31J44DI MINGW64 ~/Documents/GitHub/BTEC_640/exercises/09_21_26 (main)
$ mkdir analysis
```

1. Create your README file with a description for the content (e.g. *This directory corresponds to class exercise. September 16th, 2026*)

🔗 [Click here for a hint touch and then use nano](#)

#Paste your command here:

The structure should look like this:

:open_file_folder:btec_640

- :open_file_folder: class_exercices:
 - :open_file_folder: new_working_dir (name it following best practices)
 - :page_facing_up: README
 - :open_file_folder: input_data
 - :open_file_folder: analysis

Step 2 – Download your input data

2a. Move inside your `input_data` directoy.

:question: Which comand do you use to make sure that you are inside `input_data`?

Answer:

#Paste your command here:

```
Quant@WIN-KPSE31J44DI MINGW64 ~/Documents/GitHub/BTEC_640/exercises/09_21_26 (main)
```

```
$ cd input_data/
```

```
Quant@WIN-KPSE31J44DI MINGW64 ~/Documents/GitHub/BTEC_640/exercises/09_21_26/input_data (main)
```

```
$ pwd
```

```
/c/Users/Quant/Documents/GitHub/BTEC_640/exercises/09_21_26/input_data
```

#Paste your terminal output below:

2b. Download *E. coli* data from NCBI using curl.

```
curl -o ecoli_genomic.fna.gz "https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/GCF_000005845.2_ASM584v2_genomic.fna.gz"
curl -o ecoli_protein.faa.gz "https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/GCF_000005845.2_ASM584v2_protein.faa.gz"
curl -o ecoli_gtf.gz "https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/GCF_000005845.2_ASM584v2_genomic.gtf.gz"
curl -o ecoli_gff.gz "https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/GCF_000005845.2_ASM584v2_genomic.gff.gz"
gunzip *.gz
```

2c. Unzip the documents, **but let's do it using a loop**, no need to unzip one by one.

Let's do this step by step:

- Identify on which objects you want to execute an action or a command, in this case `gunzip`
- Find a common pattern in all the files that you just download (:bulb: You have two possible options).
- Build your order following the loop syntax.

[!TIP] Syntax: `for each` (loop over a file)

```
for FILE in "pattern"
do
  COMMAND $FILE
done
```

Note that this syntax is different from our chromosome 21 exercise, where you used `while read`.

:question: Take a moment to compare both syntaxes. Why are we not specifying the `>INPUT_FILE` here?

#Type your answer below:

Following the **for each** syntax you can use either of this options:

```
for f in ecoli*; do gunzip $f ; done
```

or

```
for f in *gz
do
  gunzip $f
done
```

*[!WARNING] :warning: :warning: :warning: Look how we are using *.
Think of the asterisk * as a "wildcard" that means "anything"
This can be super helpful but also extremely dangerous.*

Step 3 — Understanding files

Open your files with `less` and check the content, close your files typing `q`.

Notice what type of information is contained in each one of them.

FASTA:

Print the first 5 lines of both fasta files, `ecoli_genomic.fna` and `ecoli_protein.faa` and compare both files.

🔗 [Click here for a hint head -n 10 #will print the first 10 lines](#)

:question: Write the main differences that you see between both files. Describe what type of information each file has.

Type your answer here:

fna lists nucleotide sequences and is a single complete genome

faa lists amino acid sequences and is divided by respective coding genes

:question: In a FASTA file, > marks the beginning of a header line (or ID/description line) that introduces a new and independent sequence (check the sequences that you downloaded on Monday's exercise). Using grep count how many sequences each files have:

Answer:

```
grep -c ">" ecoli_protein.faa
```

returns: 4300

[!CAUTION] In coding, > normally means redirect output into a file, it is like "save and create a file". Inside quotes, ">" is just a literal character for grep to search for, not a redirection.

:warning: If you forget the quotes bash will interpret your command differently and it will create or overwrite a file instead of searching for anything. Always quote special characters like >, *, \$ when you mean them literally.

Now you know how many sequences you have in each file. Using grep, print the headers and save them in a new file name ecoli_genome_headers.txt and ecoli_proteins_headers.txt respectively (you don't need the -c flag anymore)

Paste your commands here

```
grep ">" ecoli_protein.faa > ecoli_protein_headers.txt
```

GFF and GTF

Two common file formats used in genomic data representation are Gene Transfer Format (GTF) and General Feature Format (GFF). Both are a text file, that can be read as a table.

:pencil: Exercise 3

Let's analyze and compare the annotation for **b3494** gene in *E. coli*. Both, GFF and GTF files have information about this gene. With `grep` print and save in a new file the annotation for **b3494** gene for each file. You should generate two files, named them as `b3494.gff` and `b3494.gtf` respectively, and answer the following:

1. Paste the commands that you used to print and generate the files:

2. Which are the b3494 gene coordinates in our *E. coli* genome? locate this information in both files and write down your findings:

b3494.gff file:

b3494.gtf file:

3. How many exons does this gene have? And describe where did you find this information.

b3494.gff file:

b3494.gtf file:

4. What is the Uniprot ID for this protein?

b3494.gff file: UniProtKB/Swiss-Prot:P0A8S5

b3494.gtf file: UniProtKB/Swiss-Prot:P0A8S5

5. Go to the Uniprot website (<https://www.uniprot.org/>) using the Uniprot ID that you found, look for the name and function of this protein:

FASTQ

For this section we will use a data from [The Carpentries Genomics project] (<https://datascience.si.edu/carpentries>)

which has workshops and files to practice bioinformatics. Download the fastq data from *E. coli* inside your `input_data` directory

```
curl -L -o shell_data.tar.gz "https://ndownloader.figshare.com/files/14417834"
```

:bulb: Notice that this file has a new extension type: `tar.gz`. This is a combination of a **TAR** and **GZIP**. **TAR**: Combines multiple files and directories into one single file (a "tarball") while keeping file permissions and folder structures intact, but it does not reduce the size.

GZ: Compresses the combined tar file to reduce its total size. GZ cannot compress an entire folder by itself, this is why TAR package the directories and files into a single file.

[!TIP] Syntax: `tar -[flags] <file.tar.gz>`

Flags:

`-c`: Creates a new archive

`-x`: Extract files from an archive

`-v`: Displays the archiving process

`-z`: Applies gzip compression

`-f`: Specifies the name of the archive file

Extract the files with `tar`

```
tar -xvzf <filename>
```

Take a look of the structure of the new directory that you have on your terminal. Go inside `shell_data` and do `ls` to see what is inside. Go to this directory: `shell_data/untrimmed_fastq/` :question: Open `SRR097977.fastq` file with `less` and explain what's similar and what's different between the fasta files you opened before?

Type your answer

Based on [this information](#), print the first 4 lines of `SRR097977.fastq`, describe what each line represents:

```
@SRR097977.1 209DTAAXX_Lenski2_1_7:8:3:710:178 length=36 #marks coordinates
TATTCTGCCATAATGAAATTCGCCACTTGTTAGTGT #raw sequence +SRR097977.1
209DTAAXX_Lenski2_1_7:8:3:710:178 length=36 #marks coordinates CCCCCCCCCCCCCC>CCCCC7CCCCCACA?
5A5< #gives quality report on the sequence
```